NANDI ZHANG

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EDUCATION

University of Rochester Sept. 2025 -
PhD in Computer Science
Advisor: Professor Yukang Yan

University of Calgary Sept. 2023 - Aug. 2025
MSc in Computer Science (Thesis-based)
Advisor: Professor Ryo Suzuki

Hong Kong University of Science and Technology Sept. 2018 - Aug. 2022
BSc in Data Science and Technology
Advisor: Professor Xiaojuan Ma

RESEARCH INTEREST

My research explores how humans perceive, interpret, and act mainly within **Mixed Reality**, **Human-AI Interaction**, and **Human-Robot Interaction** contexts. I build computational models that both explain and shape human perception and behavior, and I seek to advance techniques in large language models (LLMs) and computer vision to design and enable novel interactions in Mixed Reality.

PUBLICATION

[1] **Nandi Zhang**, Tongshuang Wu, Yukang Yan. *Investigating the Memory Impact of In-situ AR Text.*[1] (Submitted to **CHI'27**)

[2] Helena Qing Huang, **Nandi Zhang**, Yukang Yan. *Human-AI Collaboration for Reflective Exploration of Personal Data.*[1] (Submitted to **CHI'27**)

[3] Yu Jiang, Chun-Hao Yang, Xinning He, Xuning Hu, **Nandi Zhang**, Jrgen Steimle, and Yukang Yan. *Embodying and Becoming the Virtual Environment.*[1] (Submitted to **CHI'27**)

[4] **Nandi Zhang** and Yukang Yan. Remapping Time in Augmented Reality. (**UIST'26**)

[5] Kewen Peng, **Nandi Zhang**. Harmonia: Physiological Signal-Driven Music for Remote Relationship Maintenance. (**UIST'26** EA)
SIC Best Jury Award

[6] **Nandi Zhang**, Yukang Yan, and Ryo Suzuki. From Following to Understanding: Investigating the Role of Reflective Prompts in AR-Guided Tasks to Promote Task Understanding. (**CHI'25**)

[7] Hanfang Lyu, Xiaoyu Wang, **Nandi Zhang**, Shuai Ma, Qian Zhu, Yuhan Luo, Fu-Gee Tsung, and Xiaojuan Ma. Signaling Human Intentions to Service Robots: Understanding the Use of Social Cues during In-Person Conversations. (**CHI'25**)
Honorable Mention

[8] Aditya Gunturu, Shivesh Singh Jadon, **Nandi Zhang**, Morteza Faraji, Jarin Thundathil, Tafreed Ahmad, Wesley Willett, and Ryo Suzuki. RealitySummary: Exploring On-Demand Mixed Reality Text Summarization and Question Answering using Large Language Models. (**SUI'25**)

[1] The title of this paper has been modified from the original.

[9] **Nandi Zhang**, Nelson Wong. A Pedagogical Model for Soft-Skill Education in Computer Science: Pass/Fail Grading with Public In-Class Feedback. (**WCCCE'25**)

[10] Aditya Gunturu, Yi Wen, **Nandi Zhang**, Jarin Thundathil, Rubaiat Habib Kazi, and Ryo Suzuki. Augmented Physics: A Machine Learning-Powered Tool for Creating Interactive Physics Simulations from Static Diagrams. *In Proceedings of the Annual ACM Symposium on User Interface Software and Technology.* 2024. (**UIST'24**)
Best Paper Award

[11] Peixuan Xiong, Yukai Zhang, **Nandi Zhang**, Shihan Fu, Xin Li, Yadan Zheng, Jinni Zhou, Xiquan Hu, and Mingming Fan. To Reach the Unreachable: Exploring the Potential of VR Hand Redirection for Upper Limb Rehabilitation. *In Proceedings of the CHI Conference on Human Factors in Computing Systems.* 2024. (**CHI'24**)

WORKING EXPERIENCE

SenseTime Group Inc. Jan 2021 - Jan 2022
Research Intern

Contributed to diverse machine learning projects, addressing neural collapse in transfer learning and developing a comprehensive vision model training framework. Explored few-shot distillation techniques and implemented language models, including BERT and GPT-2. Pre-trained large-scale visio-linguistic models and worked on reinforcement learning systems like AlphaZero.

ENGINEERING EXPERIENCE

HKUST Robotics Team Feb 2019 - Aug 2022
Hardware and Software Engineer

Advisor: Professor Kam Tim Woo
Built competition robots end to end, across software, electronics, and mechanical design. Designed and built a lightweight, energy-saving robot that ran computer vision-based detection on embedded hardware and powered itself from a magnetic field through electromagnetic induction. Developed lightweight neural networks for navigation on microcontrollers (K60) and collected the training data. Designed the mechanical structure in Fusion 360 and custom PCBs in Eagle, then 3D-printed, fabricated, and soldered them.

TEACHING EXPERIENCE

CSC 442: Artificial Intelligence Fall 2026
Teaching Assistant

Instructor: Professor Daniel Gildea
Department of Computer Science, University of Rochester

CSC 242: Artificial Intelligence Spring 2026
Teaching Assistant

Instructor: Professor Adam Purtee
Department of Computer Science, University of Rochester

DATA 201: Thinking with Data Winter 2024 & Fall 2024
Head Teaching Assistant

Instructor: Professor Nelson Wong
Department of Computer Science, University of Calgary

SCIE 398: Communication for Computer Science Fall 2024
Course Development Assistant

Instructor: Professor Nelson Wong
Department of Computer Science, University of Calgary

CPSC 233: Introduction to CS for CS Majors II Fall 2024
Teaching Assistant

Instructor: Professor Jonathan Hudson
Department of Computer Science, University of Calgary

DATA 201: Thinking with Data Fall 2023
Teaching Assistant

Instructor: Professor Nelson Wong
Department of Computer Science, University of Calgary

SERVICE

Paper Review
CHI 2027 UIST 2026 CHI 2026 CHI Play 2026 CHI 2025 CSCW 2025 ISMAR 2025
Student Volunteer
UIST 2026 CHI 2024
Workshop Organization

Shuchang Xu*, Riku Arakawa*, Mina Huh*, **Nandi Zhang***, Tianyu Zhang*, Wazeer Zulfikar*, Ruei-Che Chang*, Yotam Sechayk*, Huamin Qu, Amy Pavel, Franklin Mingzhe Li, Yukang Yan, Brian A Smith, Pattie Maes.Cyber-Physical Systems for Accessibility and Ability Augmentation: Bridging Diverse Communities. (**UIST'26**)

Riku Arakawa, Franklin Mingzhe Li, **Nandi Zhang**, Mina Huh, Amy Pavel, Ryo Suzuki, Patrick Carrington, Yukang Yan. Accessible Cyber-Physical Activities. (**UIST'25**)

SKILLS

Programming Languages and Frameworks: Python (Proficient), C# (Proficient), Pytorch (Proficient), C++ (Proficient), Swift (Proficient), R (Familiar), Java (Familiar), JavaScript (Familiar), SQL and NoSQL (Familiar)
Web Technologies: React (Proficient)
Software and Tools: Unity (Proficient), Tableau (Proficient), Qualtrics (Proficient), OpenRefine (Proficient), Matlab (Familiar)
Robotics and Hardware: Fusion 360 (Familiar), Eagle (Familiar), Embedded C on microcontrollers (Familiar), Arduino (Proficient), Soldering (Proficient), 3D Printing (Familiar)
Languages: English (Fluent), Mandarin (Native)